

Classical Electrodynamics HW 3

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1 Textbook Question 3.6

Question : Two point charges q and $-q$ are located on the z axis at $z = +a$ and $z = -a$, respectively.

- a) Find the electrostatic potential as an expansion in spherical harmonics and powers of r for both $r > a$ and $r < a$.
- b) Keeping the product $qa \equiv \frac{p}{2}$ constant, take the limit of $a \rightarrow 0$ and find the potential for $r \neq 0$. This is by definition a dipole along the z axis and it's potential.

Question 3.6 Answer a): Integrate equation 82 from Appendix 4.3. The charges are located on the z axis so two of the axis have no component.

$$\Phi = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{|\vec{r} - a\hat{k}|} - \frac{q}{|\vec{r} + a\hat{k}|} \right) \quad (1)$$

Use the addition theorem where $r_{<}$ is smaller of (r, a) , and $r_{>}$ is larger of (r, a)

$$\frac{1}{|\vec{r} - \vec{r}'|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}(\theta', \phi') Y_{lm}(\theta, \phi) \quad (2)$$

Apply addition theorem to equation, using $\phi' = 0$ for both cases,

and $\theta' = \begin{cases} 0 & \text{for } r - a\hat{k} \\ \pi & \text{for } r + a\hat{k} \end{cases}$ because the charges are on the same axis.

$$\Phi = \frac{q}{4\pi\epsilon_0} \left(4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}^*(0, 0) Y_{lm}(\theta, \phi) - 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}^*(\pi, 0) Y_{lm}(\theta, \phi) \right) \quad (3)$$

Simplify

$$\Phi = \frac{q}{\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}(\theta, \phi) (Y_{lm}^*(0, 0) - Y_{lm}^*(\pi, 0)) \quad (4)$$

$$\Phi = \begin{cases} \frac{q}{\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_l}{a^{l+1}} Y_{lm}(\theta, \phi) (Y_{lm}^*(0, 0) - Y_{lm}^*(\pi, 0)) & \text{when } r < a \\ \frac{q}{\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{a^l}{r^{l+1}} Y_{lm}(\theta, \phi) (Y_{lm}^*(0, 0) - Y_{lm}^*(\pi, 0)) & \text{when } r > a \end{cases} \quad (5)$$

Question 3.6 Answer b): Simplify equation by applying azimuthally symmetric condition, all terms other than $m = 0$ vanish.

$$\Phi = \frac{q}{\epsilon_0} \sum_{l=0}^{\infty} \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}(\theta, \phi) (Y_{lm}^*(0, 0) - Y_{lm}^*(\pi, 0)) \quad (6)$$

Convert Spherical harmonics into Legendre polynomial

$$\Phi = \frac{q}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos(\theta)) (P_l(1) - P_l(-1)) \quad (7)$$

Evaluate Legendre polynomial

$$\Phi = \frac{q}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos(\theta)) (1 - (-1)^{-1}) \quad (8)$$

Simplify and alter iteration

$$\Phi = \frac{2q}{4\pi\epsilon_0} \sum_{l=0, \text{odd}}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos(\theta)) \quad (9)$$

Take the limit of $\alpha \rightarrow 0$ and apply $r \neq 0$ and $r > a$

$$\Phi = \frac{2q}{4\pi\epsilon_0} \sum_{l=0, \text{odd}}^{\infty} \frac{a^l}{r^{l+1}} P_l(\cos(\theta)) \quad (10)$$

$$\Phi = \frac{p}{4\pi\epsilon_0} \sum_{l=0, \text{odd}}^{\infty} \frac{a^{l-1}}{r^{l+1}} P_l(\cos(\theta)) \quad (11)$$

$$\Phi = \frac{p}{4\pi\epsilon_0} \left(\frac{1}{r^2} P_1(\cos(\theta)) + \frac{a^2}{r^4} P_3(\cos(\theta)) + \frac{a^4}{r^6} P_5(\cos(\theta)) + \dots \right) \quad (12)$$

Take limit $\alpha \rightarrow 0$

$$\Phi = \frac{p}{4\pi\epsilon_0} \frac{1}{r^2} \cos(\theta) \quad (13)$$

Question 3.6 Answer c): Then if dipole is surrounded by a grounded sphere shell sharing an origin, there will be additional potential from the sphere Using equation 13, add potential from sphere. The potential from the surface of a sphere can be derived from the Laplace equation in spherical coordinates (From

appendix 2, section 4.3, specifically equation 113

$$\Phi(r, \theta, \phi) = \sum_{l=0}^{\infty} (A_l r^l + B_l r^{-l-1}) P_l(\cos(\theta))$$

Where $B_l = 0$ because the region of valid solution includes the origin.

$$\Phi(r, \theta, \phi) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos(\theta)) \quad (14)$$

Add potential from sphere into equation 13

$$\Phi = \frac{p}{4\pi\epsilon_0} \frac{1}{r^2} \cos(\theta) + \sum_{l=0}^{\infty} A_l r^l P_l(\cos(\theta)) \quad (15)$$

Apply boundary condition; potential is zero at surface of sphere. Therefore, $\Phi = 0$ if $r = b$ where b is the radius of the sphere.

$$0 = \frac{p}{4\pi\epsilon_0} \frac{1}{b^2} \cos(\theta) + \sum_{l=0}^{\infty} A_l b^l P_l(\cos(\theta)) \quad (16)$$

Only $l = 1$ will be nonzero because of orthogonality

$$0 = \frac{p}{4\pi\epsilon_0} \frac{1}{b^2} \cos(\theta) + A_l b \cos(\theta) \quad (17)$$

Isolate A_l

$$A_l = -\frac{p}{4\pi\epsilon_0} \frac{1}{b^3} \quad (18)$$

Plug A_l into equation 15

$$\Phi = \frac{p}{4\pi\epsilon_0} \frac{1}{r^2} \cos(\theta) + \sum_{l=0}^{\infty} -\frac{p}{4\pi\epsilon_0} \frac{1}{b^3} r^l P_l(\cos(\theta)) \quad (19)$$

Again, Only $l = 1$ will be nonzero because of orthogonality

$$\Phi = \frac{p \cos(\theta)}{4\pi\epsilon_0 b^2} \left(\left(\frac{b}{r} \right)^2 - \frac{r}{b} \right) \quad (20)$$

2 Textbook Question 3.7

Question: Three point charges ($q, -2q, q$) are located in a straight line with separation a and with the middle charge ($-2q$) at the origin of a grounded conducting spherical shell of radius b , as indicated in the sketch.

- Write down the potential of the three charges in the absence of the grounded sphere. Find the limit form of the potential as $a \rightarrow 0$, but the product $qa^2 = Q$ remains finite. Write this latter answer in spherical coordinates.
- The presence of the grounded sphere of radius b alters the potential for $r < b$. The added potential can be viewed as caused by the surface-charge density induced on the inner surface at $r = b$ or by image charges located at $r > b$. Use linear superposition to satisfy the boundary conditions and find the potential everywhere inside the sphere for $r < a$ and $r > a$. Show that in the limit $a \rightarrow 0$,

$$\Phi(r, \theta, \phi) \rightarrow \frac{Q}{2\pi\epsilon_0 r^3} \left(1 - \frac{r^5}{b^5}\right) P_2(\cos(\theta))$$

Question 3.7, Section a) Answer: Integrate equation 82 from 4.3 for all three charges

$$\Phi = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{|\vec{r} - \alpha(\hat{z})|} + \frac{q}{|\vec{r} - \alpha(-\hat{z})|} - \frac{2q}{|\vec{r}|} \right) \quad (21)$$

Expand the first and second term using the Legendre Polynomial Expansion (Shown in Appendix 1, pg 9)

If $r > a$

$$\Phi = \frac{1}{4\pi\epsilon_0} \left(q \sum_{l=0}^{\infty} \frac{a_l}{r^{l+1}} P_l(\cos(\theta)) + q - \sum_{l=0}^{\infty} \frac{(-1)^l a^l}{r^{l+1}} P_l(\cos(\theta)) \frac{2q}{r} \right) \quad (22)$$

simplify

$$\Phi = \frac{q}{4\pi\epsilon_0} \left(\frac{-2}{r} \sum_{l=0}^{\infty} \left(1 + (-1)^l\right) \frac{a_l}{r^{l+1}} P_l(\cos(\theta)) \right) \quad (23)$$

Simplify by altering iteration

$$\Phi = \frac{q}{2\pi\epsilon_0} \left(\sum_{l=0, \text{even}}^{\infty} \frac{a_l}{r^{l+1}} P_l(\cos(\theta)) \right) \quad (24)$$

If $r > a$

$$\Phi = \frac{1}{4\pi\epsilon_0} \left(q \sum_{l=0}^{\infty} \frac{r_l}{a^{l+1}} P_l(\cos(\theta)) + q \sum_{l=0}^{\infty} \frac{(-1)^l r^l}{a^{l+1}} P_l(\cos(\theta)) - \frac{2q}{r} \right) \quad (25)$$

simplify

$$\Phi = \frac{q}{4\pi\epsilon_0} \left(\frac{-2}{r} \sum_{l=0}^{\infty} \left(1 + (-1)^l \right) \frac{a_l}{r^{l+1}} P_l(\cos(\theta)) \right) \quad (26)$$

Simplify by altering iteration

$$\Phi = \frac{q}{2\pi\epsilon_0} \left(\sum_{l=0, \text{even}}^{\infty} \frac{r_l}{a^{l+1}} \left(1 - \frac{a}{r} \delta_{l,0} \right) P_l(\cos(\theta)) \right) \quad (27)$$

As $a \rightarrow 0$, the only potential that matters is the $r > a$, and we only keep the first non-zero term

set $qa^2 = Q$

$$\Phi(r) = \frac{Q}{2\pi\epsilon_0} \left(\frac{1}{r^3} P_2(\cos(\theta)) \right) \quad (28)$$

Evaluate Legendre polynomial

$$\Phi(r) = \frac{Q}{4\pi\epsilon_0 r^3} (3\cos^2(\theta) - 1) \quad (29)$$

Question 3.7, Section b) Answer: Integrate equation 82 from Appendix 3, section 4.3 for all three charges. In this integral we can place image charges azimuthally symmetric. We can begin with equation 24 and add a element for the surface charge induced by the image charges.

$$\Phi = \sum_l A_l r^l P_l(\cos(\theta)) + \frac{q}{2\pi\epsilon_0} \left(\sum_{l=0, \text{even}}^{\infty} \frac{a_l}{r^{l+1}} P_l(\cos(\theta)) \right) \quad (30)$$

Apply boundary condition $\Phi = 0$ at surface ($r = b$)

$$0 = \sum_l A_l b^l P_l(\cos(\theta)) + \frac{q}{2\pi\epsilon_0} \left(\sum_{l=0, \text{even}}^{\infty} \frac{a_l}{b^{l+1}} P_l(\cos(\theta)) \right) \quad (31)$$

Simplify

$$0 = A_l b^l + \frac{q}{2\pi\epsilon_0} \frac{a^l}{b^{l+1}} \quad (32)$$

Isolate A_l

$$A_l = -\frac{q}{2\pi\epsilon_0} \frac{a^l}{b^{2l+1}} \quad (33)$$

Substitute back into equation

$$\Phi(r) = \sum_{l=2, \text{even}}^{\infty} \frac{q}{2\pi\epsilon_0} \left(\frac{a}{b} \right)^l \left(\frac{b^l}{r^{l+1}} - \frac{r^l}{b^{l+1}} \right) P_l(\cos(\theta)) \quad (34)$$

Since we are taking limit $a \rightarrow 0$, the $r < a$ case does not matter

Set $qa^2 = Q$, and we only need the first non-zero term

$$\Phi(r) = \frac{Q}{2\pi\epsilon_0 r^3} \left(1 - \frac{r^5}{b^5}\right) P_2(\cos(\theta)) \quad (35)$$

3 Textbook Question 3.14

Question: A line charge of length $2d$ with a total charge Q has a linear charge density varying as $(d^2 - z^2)$, where z is the distance from the midpoint. A grounded, conducting, spherical shell of inner radius $b > d$ is centered at the midpoint of the line charge.

- Find the potential everywhere inside the spherical shell as an expansion in Legendre polynomials
- Calculate the surface-charge density induced on the shell.
- Discuss your answers to parts a and b in the limit that $d \ll b$.

Question 3.14, section a) Answer: Use the Green function for the Dirichlet boundary

$$\Phi(x) = \frac{1}{4\pi\epsilon_0} \int \rho(x') G_D d^3x' - \frac{1}{4\pi} \oint (\Phi \frac{\partial G_D}{\partial n'} da' \quad (36)$$

Apply boundary condition $\Phi(x) = 0$ when $x = 2d$

$$\Phi(x) = \frac{1}{4\pi\epsilon_0} \int \rho(x') G_D d^3x' \quad (37)$$

If $r < d$ then

$$\rho(r, \theta, \phi) = \frac{3Q}{8\pi d^3} \frac{(d^2 - r^2)}{r^2} (\delta(\cos(\theta) - 1) + \delta(\cos(\theta) + 1)) \quad (38)$$

If $r > d$ then $\rho(r, \theta, \phi) = 0$

Substitute this back into equation

$$\Phi(x) = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\pi \int_0^\infty \rho(x') G_D r'^2 \sin(\theta') dr' d\theta' d\phi' \quad (39)$$

Integrate

$$\Phi(x) = \frac{3Q}{16\pi\epsilon_0 d^3} \left(\int_0^d (d^2 - r'^2) G_D(\theta' = 0) dr' + \int_0^d (d^2 - r'^2) G_D(\theta' = \pi) dr' \right) \quad (40)$$

$$G_D = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \left(\frac{r_{<}^l}{r_{>}^{l+1}} - \frac{(rr')^l}{b^{2l+1}} \right) Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) \quad (41)$$

Since there is azimuthal symmetry, all terms are removed except $m = 0$

$$G_D = 4\pi \sum_{l=0}^{\infty} \left(\frac{r_{<}^l}{r_{>}^{l+1}} - \frac{(rr')^l}{b^{2l+1}} \right) P_l(\cos(\theta')) P_l(\cos(\theta)) \quad (42)$$

Apply boundary conditions at $\theta' = 0$

$$G_D(\theta' = 0) = \sum_{l=0}^{\infty} \left(\frac{r_{<}^l}{r_{>}^{l+1}} - \frac{(rr')^l}{b^{2l+1}} \right) P_l(\cos(\theta)) \quad (43)$$

Apply boundary conditions at $\theta' = \pi$

$$G_D(\theta' = \pi) = \sum_{l=0}^{\infty} \left(\frac{r_{<}^l}{r_{>}^{l+1}} - \frac{(rr')^l}{b^{2l+1}} \right) (-1)^l P_l(\cos(\theta)) \quad (44)$$

Substitute back into equation

$$\Phi(x) = \frac{3Q}{8\pi\epsilon_0 d^3} \sum_{l=0, \text{even}}^{\infty} P_l(\cos(\theta)) \int_0^d dr' (d^2 - r'^2) \left(\frac{r_{<}^l}{r_{>}^{l+1}} - \frac{(rr')^l}{b^{2l+1}} \right) \quad (45)$$

If $r > d$ then

$$\Phi(x) = \frac{3Q}{8\pi\epsilon_0 d^3} \sum_{l=0, \text{even}}^{\infty} P_l(\cos(\theta)) \int_0^d dr' (d^2 - r'^2) \left(\frac{r'^l}{r^{l+1}} - \frac{(rr')^l}{b^{2l+1}} \right) \quad (46)$$

Evaluate integral

$$\Phi(x) = \frac{3Q}{8\pi\epsilon_0 d^3} \sum_{l=0, \text{even}}^{\infty} P_l(\cos(\theta)) \frac{d^l}{(l+3)(l+3)b^{l+1}} \left(\left(\frac{b}{r} \right)^{l+1} - \left(\frac{r}{b} \right)^l \right) \quad (47)$$

If $r < d$ then

$$\begin{aligned} \Phi(x) &= \frac{3Q}{8\pi\epsilon_0 d^3} \sum_{l=0, \text{even}}^{\infty} P_l(\cos(\theta)) \\ &\left(\int_0^r dr' (d^2 - r'^2) \left(\frac{r'^l}{r^{l+1}} - \frac{(rr')^l}{b^{2l+1}} \right) + \int_0^d dr' (d^2 - r'^2) \left(\frac{r^l}{r'^{l+1}} - \frac{(rr')^l}{b^{2l+1}} \right) \right) \end{aligned} \quad (48)$$

Evaluate Integral

$$\begin{aligned} \Phi(x) &= \frac{3Q}{8\pi\epsilon_0 d^3} \sum_{l=0, \text{even}}^{\infty} P_l(\cos(\theta)) \\ &\left(\frac{d^2(2l+1)}{l(l+1)} + \frac{r^2(2l+1)}{(2-l)(3+l)} + r^l \left(\frac{2}{d^{l-2}l(l-2)} - \frac{2d^{l+3}}{b^{2l+1}(l+3)(l+1)} \right) \right) \end{aligned} \quad (49)$$

Question 3.14, section b) Answer: Use the definition of the surface-charge density

$$\sigma = \left(-\epsilon_0 \frac{\partial \Phi}{\partial n} \right)_S \quad (50)$$

Evaluate at the inner surface $r = b$ and inward $n = -r$

$$\sigma = \left(\epsilon_0 \frac{\partial \Phi}{\partial r} \right)_{r=b} \quad (51)$$

Substitute equation in

$$\sigma = \left(\epsilon_0 \frac{\partial}{\partial r} \left(\frac{3Q}{8\pi\epsilon_0 d^3} \sum_{l=0, \text{even}}^{\infty} P_l(\cos(\theta)) \frac{d^l}{(l+3)(l+3)b^{l+1}} \left(\left(\frac{b}{r} \right)^{l+1} - \left(\frac{r}{b} \right)^l \right) \right) \right)_{r=b} \quad (52)$$

Derive and simplify

$$\sigma = -\frac{3Q}{4\pi b^2} \sum_{l=0, \text{even}}^{\infty} P_l(\cos(\theta)) \frac{(2l+1)}{(l+1)(l+3)} \left(\frac{d}{b} \right)^l \quad (53)$$

Question 3.14, section c) Answer: If $d \ll b$ then the case to evaluate is equation [46](#).

If $d \ll b$ then $\left(\frac{d}{b} \right)^l = 0$ except at $l = 0$.

$$\Phi(x) = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{b} \right) \quad (54)$$

Using equation [54](#) and the definition of surface-charge density [50](#)

$$\sigma = -\frac{Q}{4\pi b^2} \quad (55)$$

4 Appendix

4.1 Legendre Polynomial Expansion

$$\frac{1}{|\vec{r} - \vec{r}_0|} = \begin{cases} \sum_{l=0}^{\infty} \frac{r_0^l}{r^{l+1}} P_l(\cos(\theta)) & \text{if } r > r_0 \\ \sum_{l=0}^{\infty} \frac{r^l}{r_0^{l+1}} P_l(\cos(\theta)) & \text{if } r < r_0 \end{cases} \quad (56)$$

4.2 Formulation of Electric Potential

Two Electric point charges exert a force on each other.

$$\vec{F} = q\vec{E} = kq_1q_2 \frac{x_1 - x_2}{|x_1 - x_2|^3} \quad (57)$$

Electric field at point x due to a point charge

$$\vec{E} = kq_1 \frac{x - x_1}{|x - x_1|^3} \quad (58)$$

In electrostatic units (esu) $k = 1$, In SI:

$$k = \frac{1}{4\pi\epsilon_0} = 10^{-7} c^2 \quad (59)$$

Since Electrodynamics is a macroscopic theory, we will consider the "point" to

be many charged sources. Experiment shows that the fields add linearly.

$$\vec{E}(x) = k \sum_i q_i \frac{x - x_i}{|x - x_i|^3} \quad (60)$$

Consider the many charged sources to be "small" (Replace q with ρ) and "numerous" (Replace Sum with Integral) at point x' . Coulomb's law in terms of the field for a charge density. Replace singular x dimension with 3D x,y,z dimension

$$\vec{E}(x) = k \int \rho_q(x') \frac{x - x'}{|x - x'|^3} d^3 x' \quad (61)$$

A vector field can be specified completely if its divergence and curl are given everywhere in space, except up to the gradient of scalar function that satisfies the Laplace equation.

Therefore we need to specify the curl of $\vec{E}$

Starting at Coulomb's generalized integral form

$$\vec{E}(x) = k \int \rho_q(x') \frac{x - x'}{|x - x'|^3} d^3 x'$$

Using Property of Nabla (∇)

$$-\nabla\left(\frac{1}{|x - x'|}\right) = \frac{x - x'}{|x - x'|^3} \quad (62)$$

Substitute equation 72

$$\vec{E}(x) = k \int \rho_q(x') -\nabla\left(\frac{1}{|x - x'|}\right) d^3 x' \quad (63)$$

Nabla (∇) is constant; therefore,

$$\vec{E}(x) = -k \nabla \int \rho_q(x') \left(\frac{1}{|x - x'|}\right) d^3 x' \quad (64)$$

Cross nabla (∇) on both sides of equation and substitute equation

$$\nabla \times \vec{E}(x) = \nabla \times k - k \nabla \int \rho_q(x') \left(\frac{1}{|x - x'|}\right) d^3 x' \quad (65)$$

Therefore,

$$\nabla \times \vec{E} = 0 \quad (66)$$

By manipulating equation

$$\vec{E}(x) = \nabla \int \left(\frac{k \rho_q(x')}{|x - x'|}\right) d^3 x' \quad (67)$$

We can now define a portion of the equation as the scalar potential (Φ)

$$\vec{E}(x) = \nabla \Phi \quad (68)$$

Therefore the scalar potential is defined as

$$\Phi = \int \left(\frac{k\rho_q(x')}{|x - x'|} \right) d^3x' \quad (69)$$

4.3 Laplace Equation in Spherical Coordinates

The definition of the Laplace equation

$$\nabla^2 \Phi = 0 \quad (70)$$

The definition of ∇ in spherical coordinates

$$\nabla = \hat{e}_1 \frac{\partial}{\partial r} + \hat{e}_2 \frac{1}{\theta} \frac{\partial}{\partial \theta} + \hat{e}_3 \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \phi} \quad (71)$$

Where r is the radial distance from the origin, θ is the polar angle with the z-axis, ϕ is the azimuthal angle in the x-y plan relative to the x-axis

Combine the two

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} (r\Phi) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial \Phi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2 \Phi}{\partial \phi^2} = 0 \quad (72)$$

Use method of separation of variables by trying a solution of the form

$$\Phi(r, \theta, \phi) = \frac{R(r)}{r} P(\theta) Q(\phi) \quad (73)$$

Substitute equation 73 into equation 72

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} \left(r \frac{R}{r} P Q \right) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial}{\partial \theta} \left(\frac{R}{r} P Q \right) \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2}{\partial \phi^2} \left(\frac{R}{r} P Q \right) = 0 \quad (74)$$

Multiply by $\frac{r^3 \sin^2(\theta)}{R P Q}$

$$\frac{r^2 \sin^2(\theta)}{R} \frac{d^2 R}{dr^2} + \frac{\sin(\theta)}{P} \frac{d}{d\theta} \left(\sin(\theta) \frac{dP}{d\theta} \right) + \frac{1}{Q} \frac{d^2 Q}{d\phi^2} = 0 \quad (75)$$

Isolate the last term

$$\frac{r^2 \sin^2(\theta)}{R} \frac{d^2 R}{dr^2} + \frac{\sin(\theta)}{P} \frac{d}{d\theta} \left(\sin(\theta) \frac{dP}{d\theta} \right) = -\frac{1}{Q} \frac{d^2 Q}{d\phi^2} \quad (76)$$

Equate it to a constant (m)

$$\frac{1}{Q(\phi)} \frac{d^2 Q(\phi)}{d\phi^2} = -m^2 \quad (77)$$

Rearrange to get common differential equation solution

$$\frac{d^2 Q(\phi)}{d\phi^2} = -m^2 Q(\phi) \quad (78)$$

Common solutions if $m \neq 0$

$$Q(\phi) = A_m e^{im\phi} + B_m e^{-im\phi} \quad (79)$$

Common solutions if $m = 0$

$$Q(\phi) = A_m + B_m \phi \quad (80)$$

Back to Original equation 77, multiply $\sin^2(\theta)$

$$\frac{r^2}{R(r)} \frac{d^2 R(r)}{dr^2} + \frac{1}{\sin(\theta)P(\theta)} \frac{d}{d\theta} \left(\sin(\theta) \frac{dP(\theta)}{d\theta} \right) - \frac{m^2}{\sin^2(\theta)} = 0 \quad (81)$$

First and Last terms are independent so they can be set to constants

$$\frac{r^2}{R(r)} \frac{d^2 R(r)}{dr^2} - l(l+1) = 0 \quad (82)$$

Where

$$-l(l+1) = \frac{1}{\sin(\theta)P(\theta)} \frac{d}{d\theta} \left(\sin(\theta) \frac{dP(\theta)}{d\theta} \right) - \frac{m^2}{\sin^2(\theta)} \quad (83)$$

Put equation 82 into more intuitive forms

$$\frac{d^2 R(r)}{dr^2} = l(l+1) \frac{R(r)}{r^2} \quad (84)$$

Put equation 83 into more intuitive forms

$$\frac{d}{d\theta} \left(\sin(\theta) \frac{dP(\theta)}{d\theta} \right) + (l(l+1) - \frac{m^2}{\sin^2(\theta)}) \sin(\theta) P(\theta) = 0 \quad (85)$$

Reduce equation 85 to simpler form by substituting $x = \cos(\theta)$ and $\frac{d}{d\theta} = -\sqrt{1-x^2} \frac{d}{dx}$

$$\frac{d}{dx} \left((1-x^2) \frac{dP(X)}{dx} \right) + (l(l+1) - \frac{m^2}{1-x^2}) P(x) = 0 \quad (86)$$

Equation 83 by substituting $R(r) = r^\alpha$

$$\alpha(\alpha-1)r^{\alpha-2} = (l+1)r^{\alpha-2} \quad (87)$$

$$\alpha^2\alpha - 1) - l(l+1) = 0 \quad (88)$$

$$\alpha = l+1$$

$$\alpha = -l$$

General solution is

$$R(r) = A_l r^{l+1} + B_l r^{-1} \quad (89)$$

General Solution if the whole azimuthal sweep is included

$$\Phi(r, \theta, \phi) = \sum_m \sum_l (A_l r^l + B_l r^{-l-1}) (A_m d^{im\phi} + B_m e^{-im\phi} P_l^m(\cos(\theta))) \quad (90)$$

Solving for $P(\theta)$ part we consider a specific case ($m = 0$) into equation 86

$$\frac{d}{dx}((1-x^2)\frac{dP(X)}{dx}) + l(l+1)P(x) = 0 \quad (91)$$

Guess, or "Try", power series solution

$$P(x) = \sum_{j=0}^{\infty} a_j x^{j+\alpha} \quad (92)$$

$$\sum_{j=0}^{\infty} (j+\alpha)(j+\alpha-1)a_j x^{j+\alpha-2} - (j+\alpha)(j+\alpha+1)a_j x^{j+\alpha} + l(l+1)a_j x^{j+\alpha} = 0 \quad (93)$$

Consolidate

$$\sum_{j=0}^{\infty} (j+\alpha)(j+\alpha-1)a_j x^{j+\alpha-2} - \sum_{j=0}^{\infty} ((j+\alpha)(j+\alpha+1) + l(l+1))a_j x^{j+\alpha} = 0 \quad (94)$$

Remove first two powers of x then combine

$$(\alpha)(\alpha-1)a_0 x^{\alpha-2} + (1+\alpha)(\alpha)a_1 x^{\alpha-2} - \sum_{j=0}^{\infty} ((j+2+\alpha)(j+\alpha+1)a_{j+2} - ((j+\alpha)(j+\alpha+1) + l(l+1))a_j)x^{j+\alpha} = 0 \quad (95)$$

This must hold for all values of x so

$$(\alpha)(\alpha-1)a_0 = 0 \quad (96)$$

$$(1+\alpha)(\alpha)a_1 = 0 \quad (97)$$

Substitute equation

$$(j+\alpha+2)(j+\alpha+1)a_{j+2} - ((j+\alpha)(j+\alpha+1) + l(l+1))a_j = 0 \quad (98)$$

The third equation is solved to yield the recurrence relation which gives all the remaining terms from the last one. Rearrange equation

$$a_{j+2} = \frac{(j+\alpha)(j+\alpha+1) - l(l+1)}{(j+\alpha+2)(j+\alpha+1)} a_j \quad (99)$$

$a_1 = 0$; therefore, $a_{odd} = 0$

$$P_l(x) = \sum_{j=0, even}^{\infty} a_j x^{j+\alpha} \quad (100)$$

Convergest at $x = 1$

$$\frac{(j_{max} + \alpha)(j_{max} + \alpha + 1) - l(l + 1)}{(j_{max} + \alpha + 2)(j_{max} + \alpha + 1)} = 0 \quad (101)$$

If $\alpha = 0$

$$j_{max}(j_{max} + 1) - l(l + 1) = 0 \quad (102)$$

Therefore

$$j_{max} = l \quad (103)$$

Substitute equation

$$P_l(x) = \sum_{j=0, even}^l a_j x^j \quad (104)$$

Therefore

$$P_l(x) = \sum_{j=0}^{\frac{l}{2}} a_{2j} x^{2j} \quad (105)$$

Rewrite sums over all integers by doubling indices. If l is even

$$a_{j+2} = j \frac{(j + 1) - l(l + 1)}{(j + 2)(j + 1)} a_j \quad (106)$$

If $\alpha = 1$

$$(j_{max} + 2)(j_{max} + 1) - l(l + 1) = 0 \quad (107)$$

Therefore

$$j_{max} = l - 1 \quad (108)$$

Substitute equation

$$P_l(x) = \sum_{j=0, even}^{l-1} a_j x^{j+1} \quad (109)$$

Rewrite sums over all integers by doubling indices. If l is odd

$$a_{j+2} = \frac{(j + 1)(j + 2) - l(l + 1)}{(j + 3)(j + 2)} a_j \quad (110)$$

Explicitly state low values of l

$$\begin{aligned} P_0(x) &= a_0 \\ P_1(x) &= a_0 x \\ P_2(x) &= a_0 + a_2 x^2 = a_0(1 - 3x^2) \text{ where } a_2 = -3a_0 \\ P_3(x) &= a_0 x + a_2 x^3 = a_0(x - \frac{5}{3}x^3) \text{ where } a_2 = -\frac{5}{3}a_0 \end{aligned}$$

a_0 is a scale factor and it is conventional to set $P_l(1) = 1$. This conditions allows us to set the scale factor. Additionally, x is a placeholder for the more functional

$$\cos(\theta)$$

$$\begin{aligned} P_0(\cos(\theta)) &= 1 \\ P_1(\cos(\theta)) &= \cos(\theta) \\ P_2(\cos(\theta)) &= \frac{1}{2}(-1 + 3\cos^2(\theta)) \\ P_3(\cos(\theta)) &= \frac{1}{2}(-3x + \frac{5}{3}\cos^3(\theta)) \end{aligned}$$

Legendre polynomials have the following mathematical properties

$$\text{Odd or even symmetry about origin } P_l(-x) = (-1)^l P_l(x)$$

$$\text{Rodrigues' Formula } P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2 - 1)^l$$

$$\text{Recurrence relation } P_{l+1}(x) = \frac{2l+1}{l+1} P_l(x) - \frac{l}{(l+1)} P_{l-1}(x)$$

$$\text{Rodrigues' Formula } P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (P_{l+1}(x) - P_{l-1}(x))$$

$$\text{Orthogonality condition } \int_{-1}^1 P_{l'}(x) P_l(x) dx = \frac{2}{2l+1} \delta_{l'l}$$

$$\text{Orthogonality condition (Polar angle) } \int_0^\pi P_{l'}(\cos(\theta)) P_l(\cos(\theta)) \sin(\theta) d\theta = \frac{2}{2l+1} \delta_{l'l}$$

The Legendre Polynomials form a complete set of orthogonal functions on the interval $(-1, 1)$, so any function $f(x)$ can be expanded in terms of Legendre Polynomials

$$f(x) = \sum_{l=0}^{\infty} A_l P_l(x) \quad (111)$$

Where

$$A_l = \frac{2l+1}{2} \int_{-1}^1 f(x) P_l(x) dx \quad (112)$$

The general solution to the Laplace Equation in spherical coordinates for the case $m = 0$ (where the origin and non-origin are considered)

$$\Phi(r, \theta, \phi) = \sum_{l=0}^{\infty} (A_l r^l + B_l r^{-l-1}) P_l(\cos(\theta)) \quad (113)$$